# Quantum Toom's Contours

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# Introduction

Today:

- Proof.
  1. Walktrough the math

# Our Code

## Code

Consider the representation of the $d$-dimensional toric by the cayley graph of the $\left(\mathbb{Z}_n^d, +\right)$ group generated by the elements $S = \{1_i : i \in [d]\}$ [1].

In our code, bits are set on the plaquette, and the edges correspond to checks that restrict the parity of the bits on their support to be even.

[1] where $1_i$ is the vector consisting of one at the $i$th coordinate and zero elsewhere

# Local Decoder

## Positive edge

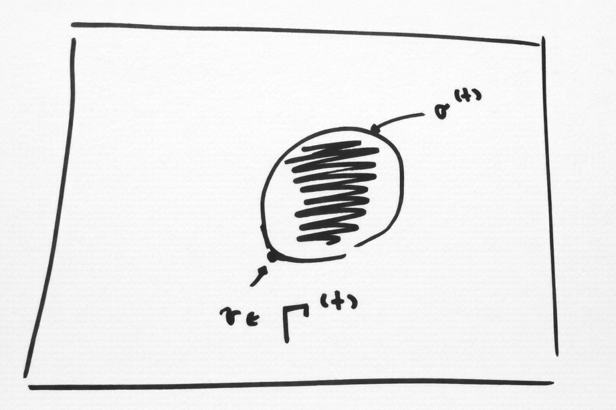
We say that an edge $\{x, y\}$ is positive relative to a plaquette $\{v, g_1 v, g_2 g_1 v, g_2 v\}$, defined by a rooted vertex $v \in \mathbb{Z}_n^d$ and the generators $g_1 \neq g_2 \in S$, if $x = v$ and $y$ is either $g_1 v$ or $g_2 v$.

## Toom's rule

We generalize Toom's rule in the following manner: Each plaquette asks if both of its positive edges point to a syndrome, namely unsatisfied. If so, then the bit flips itself; for convenience, we will think of the vertex at the end of the positive edges as the vertex which makes the decision and flips the plaquette.

# Title

Let $\sigma^{(t)}$ be a domain wall at time $t$, and let
$\Gamma^t = \{v : \exists\{v, u\} \in \sigma^t\}$.



# Title

## $u$ is added by $v$

We denote by $\phi^t$ the subsets of bits which are flipped as a result of applying the Toom's rule at time $t$, and by $\phi_v^t$ the bits which are flipped by vertex $v$.
We say that $u$ is added to $\Gamma^{t+1}$ by $v$ at time $t$, if $u \in \Gamma^{t+1}$ and $u \in V(\phi_v^t)$. When it clear form the context, we will omit the time, and say that $u$ is added by $v$.

# Title

Define the potentials functions $L_i : \mathbb{R}^d \to \mathbb{R}$ as follow:

$$L_i(\mathbf{x}) = \begin{cases} \mathbf{x}_i & i < d+1 \\ -\langle \mathbf{1}, \mathbf{x} \rangle & i = d+1 \end{cases}$$

The **Size** of a subset $S$ is defined as:

$$\mathbf{Size}(S) = \sum_i \max_{\mathbf{x} \in S} L_i(\mathbf{x})$$

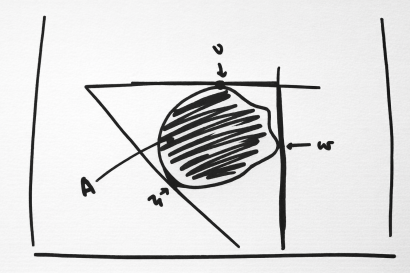



Figure: Example in 2D. $A$ is the flipped droplet, $L_x(A) = \max_{v \in \Gamma^t} v_x = w$.

# Title

## claim

Let $v \in \Gamma^t$ for any $u$ that is added by $v$, and for any $x \in [d+1]$, we have $L_x(u) \leq L_x(v) + 1$.

## proof

By definition, any $u$ that is added by $v$ is of the form $v + \sum \alpha_j 1_j$, where $\alpha_j \in \{0,1\}$ and $\sum \alpha_j \leq 2$. Thus, for any $x \in [d]$:

$$L_x(u) = \langle 1_x, v + \sum \alpha_j 1_j \rangle = \langle 1_x, v \rangle + \alpha_x \leq L_x(v) + 1$$

For $x = d+1$ we have:

$$L_x(u) = \langle -\mathbf{1}, v + \sum \alpha_j 1_j \rangle = \langle -\mathbf{1}, v \rangle - \sum_j \alpha_j \leq L_x(v)$$

# Title

claim

Let $v \in \Gamma^t$ such that $L_{d+1}(v)$ is a (local) maxima in $\Gamma^t$ then for any $u$ which is added by $v$ we have $L_{d+1}(u) \leq L_{d+1}(v) - 1$.

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## proof

We will prove that $v \notin \Gamma^{t+1}$, and then by repeating on **??** where $\sum_{\alpha_j} \geq 1$, we get the desired result. Assume, toward contradiction, that $v \in \Gamma^{t+1}$, and denote by $w$ the vertex that adds $v$. If $w \neq v$, then there is at least one coordinate $i$ for which $w_i < v_i$, thus $L_{d+1}(w) > L_{d+1}(v)$, in contradiction to $L_{d+1}(v)$ being a maximum in $\Gamma^t$. In the other case, where $v$ adds itself, it follows that $\sigma_v$ can't be decomposed into pairs of unsatisfied checks; namely, $|\sigma_v|$ is odd. We show by induction that impossible,namely that $|\sigma_v|$ is always even. The induction is over the size of the plaquettes subset that induce the boundary $\sigma$.

1. Base. Consider a boundary that induced by a single plaquette, in that case it obvious that $|\sigma_v|$ is either zero (where $v$ is not belong to the plaquette) or 2.
2. Assume for any boundary that induced by at most $k$ plaquettes, and consider a boundary $\sigma$ which is induced by $k+1$ plaquettes, name this subset $F$ and observes that $F$ can

# Title

**claim**

Let $x \in [d]$ and let $v \in \Gamma^t$ such that $L_x(v)$ is a (local) maxima in $\Gamma^t$ then for any $u$ which is added by $v$ we have $L_x(u) \leq L_x(v)$.

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## proof

Since $L_x(v)$ is maximal, $v$ doesn't see edges that goes from it at the positive $x$-direction, otherwise denote it by $e$ and denote by $w$ the second vertex at the end of $e$, $w$ has an $x$-coordinate which is greater than $v$'s $x$-coordinate by 1 and in addition $w \in \Gamma^t$ (since it lays on the support of $e$). Therefore $L_x(w) > L_x(v)$ in contradiction to $L_x(v)$ be maximal in $\Gamma^t$. Hence, any plaquette that is being flipped by $v$ lies on edges associated with $1_j$ such that $j \neq x$. Thus, any vertex $u$ that is added by $v$ has the form $v + \sum_{j\in[d]\setminus\{x\}} \alpha_j 1_j$, where $\alpha_j \in \{0,1\}$ and $\sum \alpha_j \leq 2$. So:

$$L_x(u) = \langle 1_x, v + \sum_{j\in[d]/x} \alpha_j 1_j \rangle = \langle 1_x, v\rangle = L_x(v)$$

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**claim**

Let $\Gamma^t$ be the vertices in the support of the domain wall $\sigma^t$. Denote $v_i = \arg\max_{v \in \Gamma^t} L_i(v)$. Then for any $i \in [d]$, one can find $u \in \Gamma^{t-1}$ such that $u$ adds $v_i$ and $L_i(u) \geq L_i(v_i)$. In addition, one can find $u_{d+1} \in \Gamma^{t-1}$ such that $u_{d+1}$ adds $v_{d+1}$ and $L_{d+1}(u_{d+1}) = L_{d+1}(v_{d+1}) + 1$.

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## proof

Just take $u_i$ to be any vertex that adds $v_i$. By claims **??** , **??** , and **??** , we get the claim.

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